

AI/ML intern DAY 6

TASK 1 — IMPROVING BODY MEASUREMENT ACCURACY

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1. Abstract

This task focused on improving the accuracy of AI-based body measurement estimation systems using Computer Vision and Human Pose Estimation techniques.

The project involved:

- improving landmark detection precision
- reducing measurement instability
- optimizing pose tracking
- enhancing body proportion estimation
- improving real-time AI measurement workflows

The system was tested using multiple body poses and different lighting conditions to improve overall body tracking performance and digital measurement estimation accuracy.

2. Objective

The main objective of this task was to improve the accuracy of body measurement systems using Artificial Intelligence and Computer Vision.

The task focused on:

- accurate body landmark tracking
- improved body dimension estimation
- stable real-time pose detection
- reducing landmark flickering

- improving AI body analysis
 - optimizing pose confidence levels
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3. Introduction to AI Body Measurement Systems

AI body measurement systems use Computer Vision and Pose Estimation technologies to estimate human body dimensions digitally.

The AI system analyses:

- body posture
- body landmarks
- body proportions
- shoulder width
- waist position
- body height

The detected landmarks help estimate human body measurements automatically.

These systems are commonly used in:

- virtual fitting rooms
 - fashion technology
 - fitness tracking
 - healthcare monitoring
 - AI shopping systems
 - digital tailoring systems
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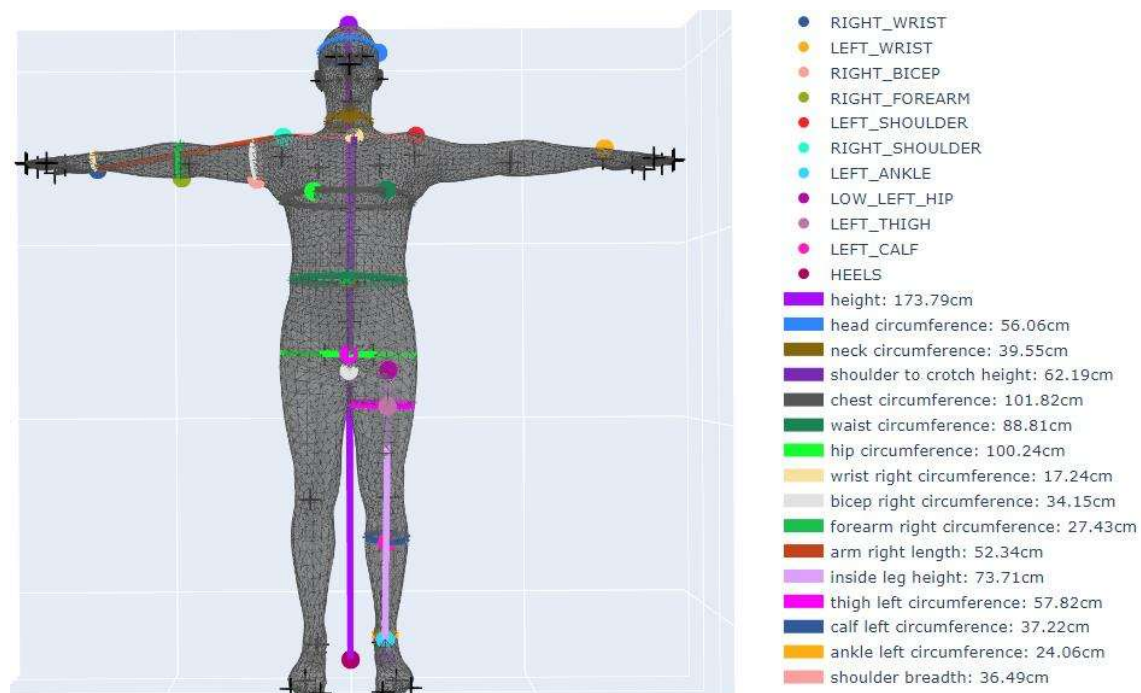


Figure 1 — AI Body Measurement Detection

4. Improved AI Measurement Workflow

The improved body measurement pipeline consists of multiple AI processing stages.

Stage 1 — Human Body Detection

The AI system first detects the human body from webcam or image input.

The system identifies:

- upper body
- lower body
- arms
- legs
- body orientation

Stage 2 — Landmark Extraction

Pose estimation algorithms detect body landmark points such as:

- shoulders
- elbows
- wrists
- hips
- knees
- ankles

These landmarks help estimate body proportions accurately.

Stage 3 — Measurement Estimation

The AI system calculates:

- shoulder width
- body height
- waist area
- torso proportions

using body landmark distance calculations.

Stage 4 — Real-Time Tracking Optimization

The system improves:

- landmark stability
 - smooth body tracking
 - measurement consistency
 - pose confidence
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Stage 5 — Final Output Rendering

The final body measurements and skeletal visualization are displayed dynamically in real time.

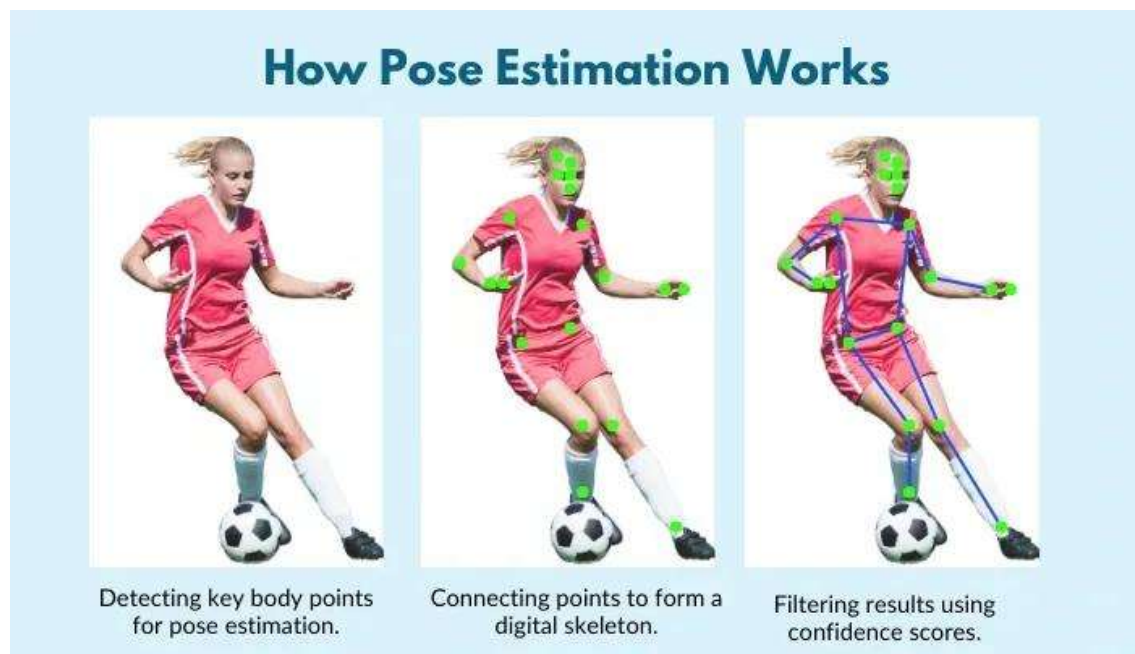


Figure 2 — Improved Real-Time Body Tracking

5. Technologies Used

Technology	Purpose
Artificial Intelligence	Smart body analysis
Computer Vision	Body tracking
MediaPipe	Pose estimation
OpenCV	Image processing
Python	System implementation
Machine Learning	Pattern analysis

6. Improvements Added Compared to Previous System

Previous System	Improved System
Basic pose detection	Advanced pose optimization
Normal landmark tracking	Stable body tracking
Low confidence levels	Increased confidence levels

Basic rendering	Smooth skeletal visualization
Minor measurement instability	Improved measurement consistency

7. Pseudo Code — AI Body Measurement Workflow

START

Capture webcam frame

Detect human body

Extract body landmarks

Estimate body dimensions

Improve landmark stability

Render body skeleton

Display body measurements

Repeat continuously for real-time tracking

END

8. Algorithm — Improved Body Measurement System

Step-by-Step Algorithm

Step 1

Initialize webcam and AI pose estimation model.

Step 2

Capture real-time video frame using OpenCV.

Step 3

Resize frame for optimized processing.

Step 4

Convert BGR image into RGB format.

Step 5

Detect body landmarks using MediaPipe Pose.

Step 6

Estimate body measurements using landmark positions.

Step 7

Apply smooth landmark tracking.

Step 8

Improve detection confidence and tracking confidence.

Step 9

Render skeletal visualization.

Step 10

Display final measurement output.

Step 11

Continue tracking dynamically in real time.

Step 12

Terminate system when exit key is pressed.

9. Improved Python Implementation

```
import cv2
import mediapipe as mp
import math

# Initialize MediaPipe Pose
mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=False,
    model_complexity=2,
    smooth_landmarks=True,
```

```
        min_detection_confidence=0.8,
        min_tracking_confidence=0.8
    )

# Start Webcam
cap = cv2.VideoCapture(0)

while True:
    success, frame = cap.read()

    if not success:
        break

    # Resize frame
    frame = cv2.resize(frame, (960, 720))

    # Convert BGR to RGB
    rgb = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

    # Process Pose Detection
    results = pose.process(rgb)

    if results.pose_landmarks:

        landmarks = results.pose_landmarks.landmark

        # Get shoulder landmarks
        left_shoulder = landmarks[11]
        right_shoulder = landmarks[12]

        # Convert normalized coordinates
        h, w, _ = frame.shape

        x1 = int(left_shoulder.x * w)
        y1 = int(left_shoulder.y * h)

        x2 = int(right_shoulder.x * w)
        y2 = int(right_shoulder.y * h)

        # Calculate shoulder width
```



```

shoulder_width = math.sqrt((x2 - x1)**2 + (y2 - y1)**2)

# Draw landmarks
mp_draw.draw_landmarks(
    frame,
    results.pose_landmarks,
    mp_pose.POSE_CONNECTIONS,
    mp_draw.DrawingSpec(color=(0,255,0), thickness=2, circle_radius=2),
    mp_draw.DrawingSpec(color=(255,0,0), thickness=2)
)

# Display shoulder width
cv2.putText(
    frame,
    f'Shoulder Width: {int(shoulder_width)} px',
    (50, 50),
    cv2.FONT_HERSHEY_SIMPLEX,
    1,
    (0, 255, 255),
    2
)

# Show Output
cv2.imshow("Improved Body Measurement System", frame)

# Exit Key
if cv2.waitKey(1) & 0xFF == ord('q'):
    break

cap.release()
cv2.destroyAllWindows()

```

10. Technical Enhancements Implemented

Enhancement	Purpose
model_complexity = 2	Improved landmark accuracy
smooth_landmarks = True	Reduced landmark flickering
High detection confidence	Better body visibility
High tracking confidence	Stable movement tracking
Frame resizing	Faster processing

11. Applications of AI Body Measurement Systems

AI body measurement systems are used in:

- virtual try-on systems
 - AI fashion technology
 - healthcare monitoring
 - fitness applications
 - sports analytics
 - digital tailoring systems
 - body posture analysis
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12. Challenges Faced

Challenge	Solution
Poor lighting	Improved lighting setup
Fast body movement	Increased tracking confidence
Landmark instability	Enabled smooth landmark tracking
Frame lag	Optimized frame resizing

13. Knowledge and Skills Acquired

During this task, the following concepts were understood:

- advanced pose estimation
 - body measurement analysis
 - landmark optimization
 - AI body tracking
 - computer vision workflows
 - real-time AI processing
 - skeletal visualization systems
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14. Conclusion

The improved body measurement system successfully enhanced real-time body tracking accuracy and body dimension estimation using Artificial Intelligence and Computer Vision technologies.

The system provided:

- stable landmark tracking
- improved body measurement estimation
- better pose confidence
- smoother skeletal rendering
- optimized real-time performance

This task improved understanding of:

- AI pose estimation
- body measurement systems
- landmark tracking optimization
- real-time computer vision processing
- AI body analysis workflows